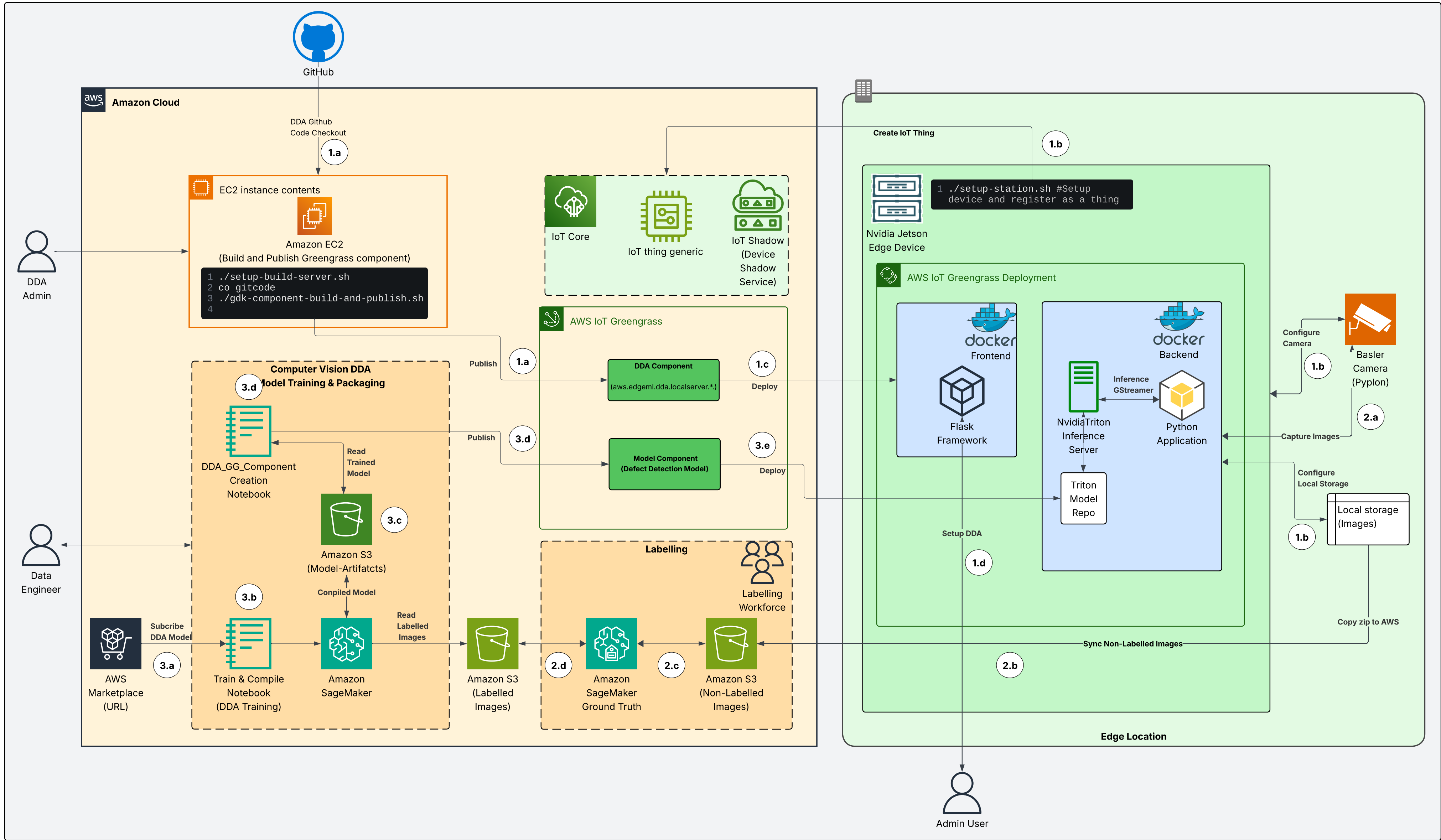


Defect Detetection Application (DDA) Build and Deploy Steps



LEGEND:

1. Device Setup & IoT Configuration:

- **1.a** DDA Admin checks out DDA GitHub repo (code + configuration) on build server (Ec2 instance). Then builds and publishes the DDA application component in Greengrass(GG) component repository
- **1.b** Admin User configures Basler Camera (Pylon), sets up local storage for images, and registers edge device (station) as IoT Thing via setup-station.sh.
- **1.c** Once the station is setup, use the GG Cli or AWS console to publish the DDA component to the edge device
- **1.d** Admin sets up DDA application by logging into browser application.
-

2. Data Acquisition & Labeling:

- **2.a** Basler Camera captures images using the DDA application.
- **2.b** Non-labeled images synced from edge device → Amazon S3 (Non-Labeled Images) by zipping the images.
- **2.c** Non-labeled images ingested into **Amazon SageMaker Ground Truth** for human/auto labeling.
- **2.d** Labeled images stored in **Amazon S3 (Labelled Images)**.
-

3. Model Training & Packaging:

- **3.a** Data Engineer subscribes to DDA Base Model from **AWS Marketplace**.
- **3.b** Amazon SageMaker Training Notebook trains and compiles the Computer Vision DDA model using labeled images.
- **3.c** Trained model artifacts are stored in **Amazon S3 (Model Artifacts)**.
- **3.d** DDA_GG Component Creation Notebook packages models → publishes components to AWS IoT Greengrass Component registry
- **3.e** Use the GG Cli or AWS console to publish the Model component to the edge device